Evolution and Reinforcement Learning in Changing Environments

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Abstract

We compare ecological neuroevolution (OE), evolution strategies (ES) and proximal policy optimization (PPO) in a simulated foraging world. Six separately tuned fixed- and varied-environment configurations train for the same nominal agent-step budget, with ten runs each. We measure food acquisition before and during adaptation to thirty test environments and population persistence without replacement support. Across twenty shifted environments, OE performs below PPO with domain randomization (PPO-DR): the difference in interquartile mean of normalized adaptation area under the curve is -0.5592 (nominal 95% bootstrap interval $[-0.6088, -0.5196]$). Varied OE also performs below OE-fixed. ES-DR slightly exceeds ES-fixed, whereas PPO-DR's small advantage over PPO-fixed is nonsignificant in the prespecified rank test. OE has poorer population persistence than PPO-DR and depends heavily on its population floor: externally supplied replacements account for over 99% of births during resource-scarce adaptation.

1 Introduction

Milano et al. (2019) found that environmental variation across generations improved evolved controllers, with the best performance at a moderate rate of change. In ecological neuroevolution, controllers are selected through survival and reproduction. An agent must acquire enough food to remain alive and produce offspring, so this selection pressure may favor different controllers from direct optimization of food intake.

We compare ecological neuroevolution (OE), evolution strategies (ES) and proximal policy optimization (PPO) in a continuous two-dimensional foraging world. For each method, we train controllers in either a fixed environment or a sequence of varied environments. We then measure food acquisition before and during adaptation to new environments, and population persistence in a separate assay. OE maintains an interacting population with energy-dependent death and reproduction and a minimum population supplied by replacements. ES and PPO optimize food acquisition and deploy copies of a single controller.

We predicted that varied-environment OE would outperform PPO with domain randomization during adaptation. Instead, OE starts below PPO-DR and remains below it throughout the measured interval. A supporting experiment with inherited traits and fixed behavioral rules examines search objectives, mutation and cloning (Appendix D).

2 Related work

Milano and Nolfi (2024) compared ES and PPO across tasks and treatments of environmental variation. Our experiment adds ecological selection to this comparison in a shared foraging task.

Table 1: Training ranges and test suites. Each test suite contains ten environments; layout parameters apply only to the corresponding layout.

Parameter	Varied training / T1	T2	T3
Arena side L	40–80	30–40	40–80
Food items F	20–60	10–20	20–60
Layouts	Uniform, clusters, patches	Clusters	Uniform, clusters, patches
Cluster count k	2–5	2–3	2–5
Cluster scale σ	3–8	2–4	3–8
Patch radius	4–10	Not used	4–10
Regrowth probability	0.01–0.05	0.005–0.01	0.01–0.05
Food drift	0	0	0.05–0.3
Season period (steps)	None	None	100–300

Hamon et al. (2023) studied persistent neural-agent populations with physiology-dependent reproduction and death. Nisioti and Moulin-Frier (2022) used a minimal ecological model and ablations to distinguish fitness-based selection from niche-limited competition. Our implementation couples survival and reproduction to the behavior of feedforward neural controllers.

Prior comparisons examine curricula (Nisioti et al., 2025), sequential changes and forgetting (Nisioti et al., 2026), and update and inheritance regimes in ecological evaluations (Ge and Cheng, 2026). Here, adaptation begins independently from the trained snapshot in each test environment and permits further parameter updates.

3 Experimental design

3.1 Foraging environment and controllers

Agents move in a square arena with walls and regenerating food. Their observations contain 30 values: eight directional sectors for each of food, other agents and walls; current energy; the previous two-dimensional action; and three body traits. A feedforward controller has two hidden layers of 32 tanh units and two outputs, controlling turn and forward speed. Every method uses this actor architecture. Maximum speed, size and sensing range are fixed at 1, 1 and 10, respectively.

Energy decreases with time, movement and sensing. A food item supplies 40 energy units; an agent starts with 100 and dies when its energy reaches zero. In ecological mode, agents can reproduce above an energy threshold, paying an energy cost for each offspring; they also die at a maximum age. Appendix A gives the full settings. Episodic competence evaluation uses the same movement, sensing and food dynamics, without reproduction.

Varied training samples arena size, food supply, layout and regrowth from the ranges in Table 1. The fixed environment has side length 60, 40 food items, a uniform layout and regrowth probability 0.03. Validation uses 20 sampled training-range environments and four corner cases. We fix thirty test environments before final evaluation: ten within the training ranges (T1), ten smaller, resource-scarce environments with clustered food (T2), and ten with food drift and seasonal relocation (T3). T3 retains the training ranges for the remaining parameters.

Table 2: Training and transfer procedures. Each row has fixed and varied variants, separately tuned. Body traits and actor architecture are shared.

Procedure	Training signal	Competence deployment	State at adaptation start
OE	Energy-dependent survival/reproduction plus floor replacements	16 sampled population members	Genomes/alive mask retained; world state and energy reset
ES	Rank-shaped episodic food score	16 copies of mean actor	Mean solution retained; fresh strategy/optimizer
PPO	Food reward and clipped policy objective	16 copies of actor	Actor/critic and reward statistics retained; fresh Adam/rollout state

3.2 Training procedures

Ecological neuroevolution. OE starts with 32 agents and permits at most 128. Each agent has its own controller, and offspring inherit the parent's weights with Gaussian perturbations. In varied OE, the environmental parameters change every 2,000 world steps while the population continues. OE-fixed retains the canonical environment. During adaptation, either variant continues evolving in the assigned test environment.

Both variants maintain a population floor of 16. After a step below that floor, missing agents are replaced with mutated offspring of uniformly sampled surviving, non-newborn agents; if none are available, agents alive at the start of the step supply the parents. Replacements receive birth energy and a new position. We count natural and floor births separately.

Evolution strategies. ES optimizes the shared actor weights using `Open_ES` in evosax (Salimans et al., 2017; Lange, 2023), with antithetic perturbations and centered-rank fitness shaping. ES-DR samples training environments; ES-fixed uses the canonical environment. Evaluation uses the strategy mean.

Each candidate is evaluated with 16 policy copies in 500-step worlds without births. Its score is food per initial agent, averaged over its configured episodes. Candidates in a generation share environment draws and reset/step random keys; ES-DR redraws the episode environments each generation.

Proximal policy optimization. PPO uses a clipped policy-gradient objective (Schulman et al., 2017), a separate value network, and a shared policy across agents in 64 parallel episodic worlds. Food acquisition supplies the reward. PPO-DR samples environment parameters at episode resets; PPO-fixed uses the canonical environment. Training actions are stochastic; competence evaluation uses deterministic actions. We adapt PureJaxRL's continuous-action PPO (Appendix A).

Each world contains 16 agents and restarts at 500 steps or when all agents die. An agent's death terminates its trajectory. Steps beginning with a dead agent are excluded from losses and advantage normalization; when enabled, reward-normalization statistics are collected over living agents.

Table 2 summarizes deployment and transfer.

3.3 Tuning, budgets and evaluation

We tune each variant separately with eight random-search trials, each using a distinct seed and 5×10^7 nominal living-agent steps. We select the configuration with the highest mean zero-shot food acquisition above the random-policy score across the 24 validation environments, then train it with ten final seeds, 5000–5009, at 2×10^8 nominal living-agent steps per run. Selection thus uses zero-shot performance, whereas the primary outcome measures performance during adaptation. Fixed-versus-varied comparisons also include differences in selected hyperparameters (Appendix A).

OE, OE-fixed, PPO-DR, PPO-fixed and ES-DR form the initial comparison. We added ES-fixed after inspecting these results, before its own tuning and final runs.

One living-agent step counts an agent alive at the beginning of a simulated step. Equal agent-step budgets do not equate wall time, floating-point operations or optimizer updates. Updates finish in indivisible units and may cross a checkpoint or stopping target; we analyze both nominal and actual step counts.

Competence is total food consumed divided by the 16 initial agents in a 500-step episode, averaged across eight evaluation episodes. OE samples population members afresh for each episode, without replacement when at least 16 are alive; ES and PPO use 16 policy copies. Episode and reset seeds are shared across methods, although the random-action reference does not share every subsequent world-step random draw with learned policies.

Adaptation starts independently from the final trained snapshot in each test environment and receives 2×10^7 nominal living-agent steps. We measure competence before adaptation and at ten subsequent checkpoints. Table 2 and Appendix A specify the transferred state.

Persistence is evaluated separately, before adaptation. Thirty-two sampled OE agents or policy copies enter ecological mode for 5,000 world steps, with reproduction but no mutation and no population floor. We record extinction and steps survived.

3.4 Outcomes and statistical analysis

Let $c_{mre}(t)$ be competence for method m, run r and environment e at normalized adaptation checkpoint $t \in \{0, 0.1, \ldots, 1\}$. We use the prespecified transformation

$$z_{mre}(t) = \frac{c_{mre}(t) - q_e}{b_e - q_e}, \qquad A_{mre} = \frac{1}{10} \sum_{j=1}^{10} \frac{z_{mre}(t_{j-1}) + z_{mre}(t_j)}{2}, \tag{1}$$

where q_e is the random-policy score and b_e is the largest score across the five initial configurations, all final runs and all eleven checkpoints in that environment. Every environment has $b_e > q_e$. ES-fixed uses the same anchors.

The primary effect is

$$D = \mathrm{IQM}\{A_{\mathrm{OE},r,e} : e \in T2 \cup T3\} - \mathrm{IQM}\{A_{\text{PPO-DR},r,e} : e \in T2 \cup T3\}, \tag{2}$$

where the interquartile mean (IQM) averages the middle half of the pooled run–environment values. We bootstrap complete training runs, keeping their environment outcomes together, with 50,000 draws and statistics seed 4800 (Agarwal et al., 2021; Henderson et al., 2018). The prespecified margin is 0.05 on this reference scale. AUC measures performance over the adaptation interval, including starting competence.

The seven initial secondary comparisons use two-sided Mann–Whitney tests on per-run environment means, with Holm adjustment within the family. The ES comparisons form a separate

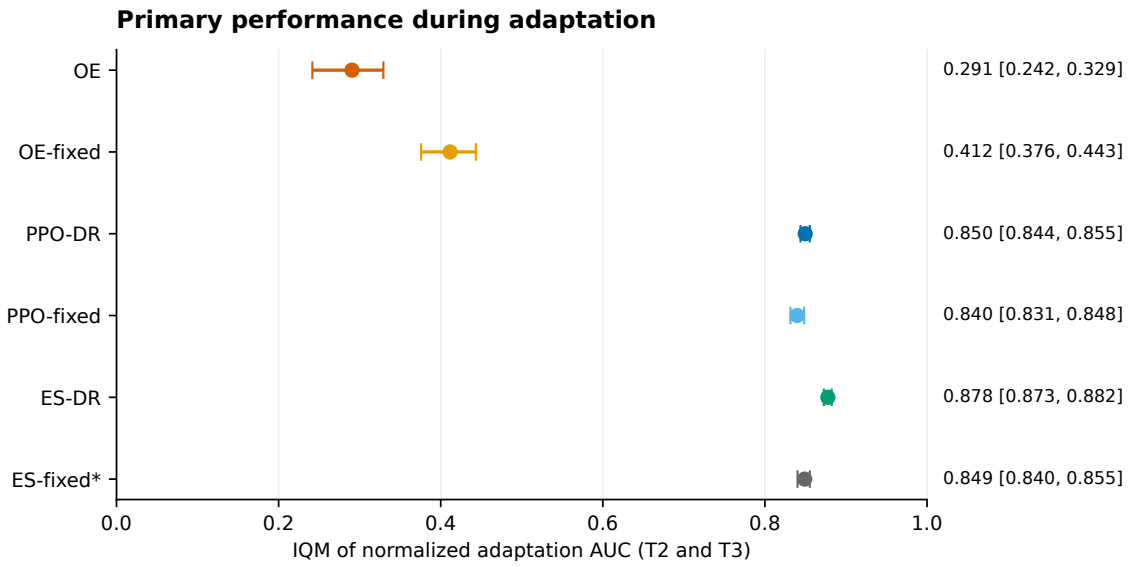


Figure 1: Performance during adaptation on T2 and T3. Points show pooled IQMs of normalized AUC; bars are nominal 95% whole-run percentile bootstrap intervals from 50,000 draws. Each procedure has ten training runs evaluated on twenty fixed environments. Asterisk: ES-fixed was added after the core experiment.

family. The rank tests compare distributions of run means; the effect estimates compare pooled IQMs. Appendix B gives the comparisons and decision rules.

Bootstrap intervals hold the normalization anchors and thirty test environments fixed. They exclude uncertainty from environment and episode sampling, tuning and anchor selection. Because the maximum anchors come from the same runs, interval coverage is uncertain. We examine anchor re-estimation and paired resampling of corresponding seeds in Appendix C.

4 Results

4.1 OE scores below PPO-DR during adaptation

Across T2 and T3, OE has an adaptation AUC IQM of 0.2905, compared with 0.8497 for PPO-DR. Their difference is -0.5592 (nominal 95% interval $[-0.6088, -0.5196]$). OE also performs below ES-DR, with an IQM difference of -0.5871 (Holm-adjusted $p = 0.00110$). Figure 1 compares all six configurations.

OE begins below PPO-DR and remains below it during adaptation (Figure 3). The zero-shot IQM difference on T2/T3 is -0.4648; at the final checkpoint it is -0.6035 (both adjusted $p = 0.00110$). In T2, OE's mean normalized competence declines from 0.4383 to 0.2338, while PPO-DR changes from 0.7244 to 0.7478. T3 shows a smaller decline for OE, from 0.2966 to 0.2655.

On the in-range T1 suite, the OE–PPO-DR difference in adaptation AUC is -0.5371 (nominal 95% interval $[-0.5951, -0.4799]$).

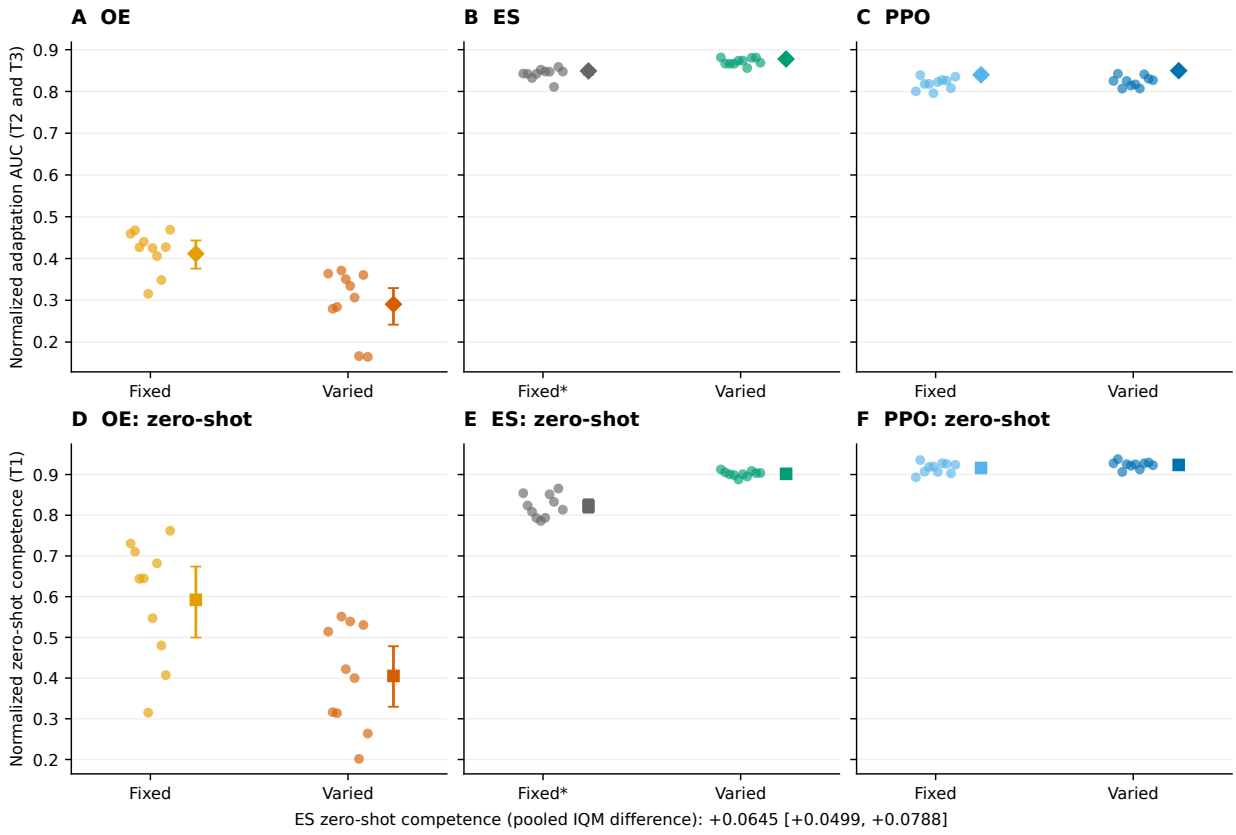


Figure 2: Fixed and varied configurations. Upper panels: T2/T3 AUC, with run means (circles), pooled IQMs (diamonds) and nominal 95% whole-run intervals. Lower panels: T1 zero-shot run means and their mean (squares), with descriptive 95% intervals from 20,000 whole-run bootstrap draws. The ES zero-shot contrast uses pooled IQMs; lower OE/PPO panels are descriptive. Each configuration was tuned separately. Asterisk: ES-fixed extension.

4.2 Fixed and varied training

Varied OE performs below OE-fixed on T2/T3 AUC: difference -0.1210, interval $[-0.1799, -0.0680]$ (adjusted $p = 0.00440$). PPO-DR has no significant advantage over PPO-fixed in the rank test (adjusted $p = 0.6232$); the IQM difference is 0.0099, interval $[-0.0004, 0.0201]$.

ES-DR exceeds ES-fixed in T2/T3 adaptation AUC by 0.0286 (nominal 95% interval $[0.0203, 0.0384]$), below the 0.05 margin. Its T1 zero-shot advantage is 0.0645 (interval $[0.0499, 0.0788]$). Both rank tests give adjusted $p = 0.000365$.

4.3 Population persistence and support

OE has higher extinction than PPO-DR in the persistence assay. Across T2/T3, the difference in IQMs of per-run extinction fractions is 0.3750, interval $[0.2500, 0.4833]$ (adjusted $p = 0.00110$). Extinction is frequent across all six configurations in resource-scarce T2, including those with high normalized competence (Table 3).

Table 3: Extinction frequencies in the 5,000-step persistence assay without floor support. Each cell pools ten training runs evaluated in ten environments. Asterisk: ES-fixed extension.

Procedure	T1	T2	T3
OE	0.48	0.94	0.76
OE-fixed	0.30	0.93	0.48
PPO-DR	0.03	0.83	0.15
PPO-fixed	0.10	0.95	0.14
ES-DR	0.03	0.89	0.22
ES-fixed*	0.11	0.87	0.21

Table 4: Share of new agents supplied by population-floor replacements during adaptation. Fractions pool event counts over ten runs and ten environments per suite.

Suite	OE	OE-fixed
T1	64.35%	59.29%
T2	99.53%	99.60%
T3	78.90%	71.87%

The floor supplies approximately 58% of training births for OE and 59% for OE-fixed, averaged across runs. During T2 adaptation (Table 4), the pooled shares rise to 99.53% and 99.60%, respectively.

4.4 Learning curves

Between the first validation checkpoint near 20 million steps and the end of training at 200 million steps, OE's mean score above random increases from 9.4818 to 10.6102; six of ten runs improve (Figure 3). Appendix C reports three exploratory runs extended to one billion steps.

The OE–PPO-DR contrast remains negative under paired-seed resampling, re-estimation of normalization anchors and integration over actual step counts (Appendix C).

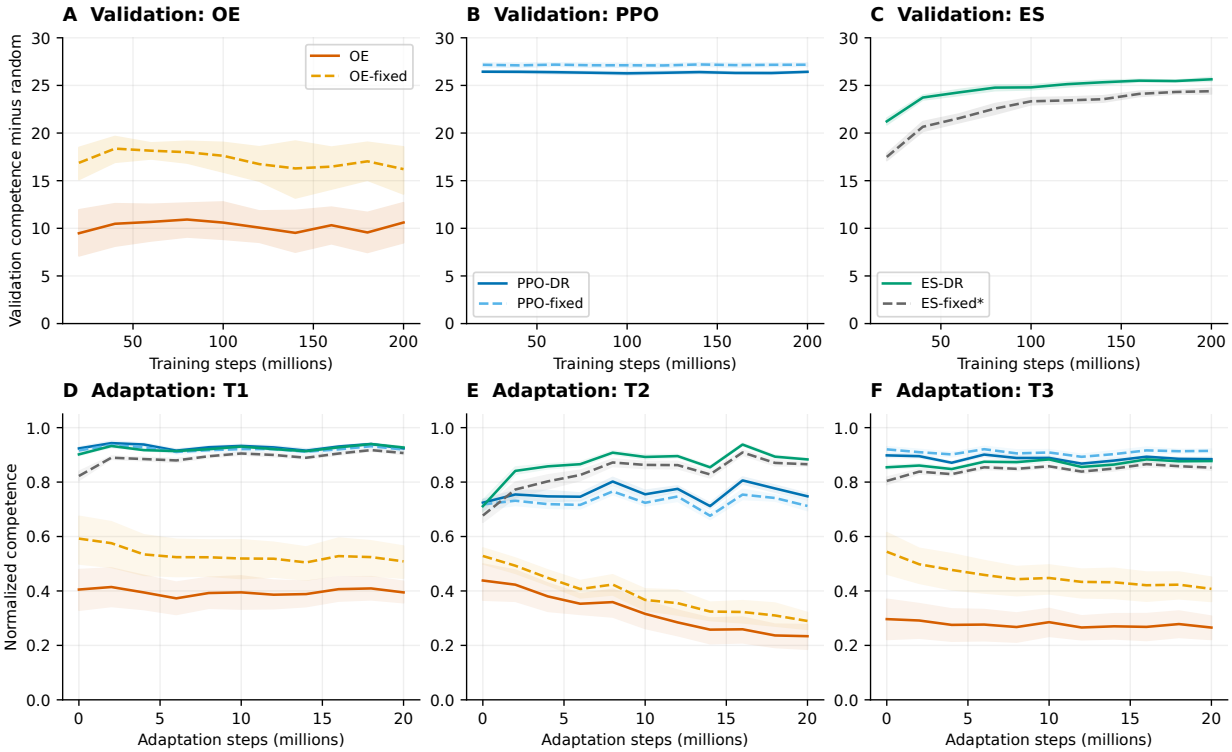


Figure 3: Training and adaptation. Upper panels: validation competence above the random reference. Lower panels: normalized test competence during adaptation. Lines average ten runs; shading gives pointwise 95% whole-run percentile intervals from 20,000 draws. Axes use nominal living-agent steps. Training measurements begin near 20 million steps.

5 Discussion

OE performs below PPO-DR in both food acquisition and population persistence. The ecological objective, population floor and heterogeneous deployment are possible explanations for this deficit. Comparisons that match the objective and deployment population, then vary replacement support, would help distinguish their contributions.

In resource-scarce T2, even policies with high normalized food scores frequently go extinct. Normalization compares intake with that of other policies; it does not measure whether intake covers the energy costs of survival and reproduction. The OE floor supplies both replacement agents and energy, allowing the population to continue when natural reproduction is insufficient.

The ES zero-shot advantage under varied training agrees in direction with Milano et al. (2019), although its adaptation advantage is smaller. Their study found the best performance at a moderate rate of change. We used one change rate for OE and therefore cannot determine whether a different rate would improve its performance. A rate sweep with common hyperparameters would test this possibility.

Data and code availability

Code, data and manuscript source are available at `https://github.com/Guntikiwritescode/ecological-neuroevolution`. Reproduction instructions are in `REPRODUCE.md`.

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A Implementation and selected settings

The actor has 2,114 parameters. Initial weights are orthogonal, with gain $\sqrt{2}$ in hidden layers and 0.01 in the action layer; biases start at zero. The world maps the outputs through tanh to a turn bounded by $\pi/4$ per step and a speed between zero and the body's maximum. Observations include clipped energy divided by birth energy and scaled body traits.

The per-step energy cost is $0.5 + s^3v^2 + 0.05r$, where s is size, v is commanded speed after action squashing but before position clipping at a wall, and r is sensing range. With the default body it is $1 + v^2$. Ecological reproduction requires energy at least 200, costs the parent 120 and supplies the offspring with 100. Maximum age is 1,000 steps and the offspring placement offset is 1.

Within a step, agents move, consume reachable food, pay costs, die and reproduce; food dynamics then update. Random priority resolves contested food. Births require free slots, with eligible parents processed in random order, at most one child per parent per step, and payment only for successful births. At a varied-OE environment change, positions are clipped to the new arena as needed, while energy, age and genomes persist. Food is reinitialized under the new parameters.

Both PPO variants use 32 minibatches, GAE parameter 0.95, clipping parameter 0.2, value-loss coefficient 0.5, maximum gradient norm 0.5, initial log standard deviation zero and no learning-rate annealing. Table 5 gives the selected hyperparameters. Adaptation retains PPO's actor parameters, critic parameters, log standard deviation and reward-normalization state, while resetting its return accumulator, optimizer, environment state, counters and random stream. OE retains controller genomes and the alive mask, while resetting positions, headings, energy, age, actions and counters. ES retains its mean solution while restarting its optimizer, strategy and generation counter.

The PPO critic has the same two hidden layers as the actor and a scalar output. Exploration uses Gaussian actions with learned, state-independent log standard deviation. We use PureJaxRL commit `31756b1` and evosax version 0.3.2.

Table 5: Selected hyperparameters, rounded to six significant digits. Exact values are in the accompanying tuning CSV. Asterisk: ES-fixed extension.

Procedure	Settings
OE	Mutation SD = 0.0341186
OE-fixed	Mutation SD = 0.00507475
PPO-DR	Entropy coefficient = 0; Discount = 0.99; Learning rate = 0.000180353; Reward normalization = on; Rollout length = 64; Update epochs = 8
PPO-fixed	Entropy coefficient = 0; Discount = 0.95; Learning rate = 0.000706297; Reward normalization = off; Rollout length = 128; Update epochs = 8
ES-DR	Episodes per candidate = 1; Learning rate = 0.0073746; Population size = 64; Perturbation SD = 0.089879
ES-fixed*	Episodes per candidate = 4; Learning rate = 0.0117805; Population size = 32; Perturbation SD = 0.0441052

For PPO tuning, we sampled learning rate log-uniformly from 10^{-4} to 10^{-3}, rollout length from $\{32, 64, 128\}$, update epochs from $\{2, 4, 8\}$, discount from $\{0.95, 0.99, 0.995\}$, entropy coefficient from $\{0, 0.001, 0.01\}$ and reward normalization on or off. For ES, we sampled perturbation scale from $[0.01, 0.1]$ and learning rate from $[0.003, 0.03]$, both log-uniformly, population size from $\{32, 64, 128\}$ and episodes per candidate from $\{1, 2, 4\}$. For OE, we sampled weight-mutation standard deviation

log-uniformly from $[0.005, 0.05]$, keeping the population floor and change interval fixed. Score ties favored the lower trial index.

Validation environments use seeds 4900–4919 plus four corner cases; T1, T2 and T3 use seeds 5900–5909, 5910–5919 and 5920–5929. Competence episodes use environment-specific evaluation seeds and checkpoint indices. Random streams are separated by purpose; shared seeds can couple corresponding runs across methods. The evaluation data include each environment's parameter values.

B Additional neural comparisons

The OE–PPO-DR and ES-DR–ES-fixed adaptation comparisons use an interval rule with a margin of 0.05. An interval wholly above zero supports the predicted direction; one wholly below zero opposes it. An interval between zero and $+0.05$, or between -0.05 and zero, is also below the practical margin. If neither directional condition holds, an interval contained in $[-0.05, +0.05]$ is labeled no meaningful difference; any other interval is inconclusive. Table 6 combines these intervals with the secondary rank-test results.

Table 6: Neural comparisons. Differences are pooled IQMs; extinction uses IQMs of per-run extinction fractions. Intervals are nominal 95% whole-run intervals. Adjusted p values compare per-run summaries, with separate Holm correction for the seven initial secondary comparisons and the two ES comparisons. The primary OE–PPO-DR AUC comparison uses the interval rule alone. It places OE below PPO-DR by more than 0.05; ES-DR exceeds ES-fixed by less than 0.05.

Comparison	Outcome	Difference [95% interval]	Adjusted p
OE minus PPO-DR	T2/T3 AUC	-0.5592 $[-0.6088, -0.5196]$	–
OE minus ES-DR	T2/T3 AUC	-0.5871 $[-0.6364, -0.5475]$	0.0011
OE minus OE-fixed	T2/T3 AUC	-0.1210 $[-0.1799, -0.0680]$	0.0044
PPO-DR minus PPO-fixed	T2/T3 AUC	$+0.0099$ $[-0.0004, 0.0201]$	0.6232
OE minus PPO-DR	T2/T3 zero-shot	-0.4648 $[-0.5570, -0.4073]$	0.0011
OE minus PPO-DR	T2/T3 final	-0.6035 $[-0.6486, -0.5643]$	0.0011
OE minus PPO-DR	T1 AUC	-0.5371 $[-0.5951, -0.4799]$	0.0011
OE minus PPO-DR	T2/T3 extinction	$+0.3750$ $[0.2500, 0.4833]$	0.0011
ES-DR minus ES-fixed	T2/T3 AUC	$+0.0286$ $[0.0203, 0.0384]$	3.65×10^{-4}
ES-DR minus ES-fixed	T1 zero-shot	$+0.0645$ $[0.0499, 0.0788]$	3.65×10^{-4}

The Mann–Whitney probability summary is $U/(n_X n_Y)$, with ties contributing one half. Extinction comparisons use per-run extinction fractions across T2/T3. The OE–PPO-DR comparisons of T2/T3 zero-shot competence and T1 AUC were nondirectional. The remaining secondary comparisons predicted an OE advantage over ES-DR and OE-fixed in T2/T3 AUC, an OE advantage over PPO-DR in final competence and persistence, a PPO-DR advantage over PPO-fixed in AUC, and ES-DR advantages over ES-fixed in AUC and T1 zero-shot competence.

Table 7: Fixed–varied contrasts for OE/PPO and OE/ES. The OE/ES interaction is the ES varied-minus-fixed difference minus the OE difference.

Methods	Quantity	Estimate	Nominal 95% interval
OE/PPO	$\Delta = \lvert V\rvert - \lvert M\rvert$	-0.4382	[-0.4743, -0.4056]
OE/PPO	I	-0.1310	[-0.1907, -0.0764]
OE/PPO	M	-0.4937	[-0.5240, -0.4671]
OE/PPO	V	-0.0556	[-0.0852, -0.0282]
OE/ES	I	+0.1496	[0.0958, 0.2091]
OE/ES	ES-DR minus ES-fixed	+0.0286	[0.0203, 0.0385]
OE/ES	OE minus OE-fixed	-0.1210	[-0.1797, -0.0678]

For the OE/PPO comparison, write $q_{ov}, q_{of}, q_{pv}, q_{pf}$ for pooled AUC IQMs of OE, OE-fixed, PPO-DR and PPO-fixed. The prespecified summaries are

$$V = \tfrac{1}{2}[(q_{ov} - q_{of}) + (q_{pv} - q_{pf})], \tag{3}$$

$$M = \tfrac{1}{2}[(q_{ov} - q_{pv}) + (q_{of} - q_{pf})], \tag{4}$$

$$I = (q_{ov} - q_{of}) - (q_{pv} - q_{pf}), \qquad \Delta = |V| - |M|. \tag{5}$$

Here V averages the two varied-minus-fixed contrasts, M averages the two OE-minus-PPO contrasts, and I measures their interaction (Table 7). The absolute-value contrast Δ is unstable when a component is close to zero.

The exploratory OE/ES comparison subtracts the OE varied-minus-fixed contrast from the ES contrast: difference 0.1496, interval $[0.0958, 0.2091]$. Its component intervals use joint four-method resampling, whereas the individual comparisons use two-method resampling.

Table 8: Mean normalized performance, averaging environments within runs and then runs. Asterisk: ES-fixed extension.

Procedure	Suite	Zero-shot	Final	AUC
OE	T1	0.4052	0.3946	0.3960
OE	T2	0.4383	0.2338	0.3205
OE	T3	0.2966	0.2655	0.2759
OE-fixed	T1	0.5923	0.5088	0.5303
OE-fixed	T2	0.5285	0.2897	0.3860
OE-fixed	T3	0.5441	0.4076	0.4510
PPO-DR	T1	0.9236	0.9235	0.9295
PPO-DR	T2	0.7244	0.7478	0.7611
PPO-DR	T3	0.8981	0.8840	0.8861
PPO-fixed	T1	0.9161	0.9196	0.9216
PPO-fixed	T2	0.7181	0.7123	0.7290
PPO-fixed	T3	0.9202	0.9147	0.9089
ES-DR	T1	0.9017	0.9270	0.9231
ES-DR	T2	0.7114	0.8833	0.8743
ES-DR	T3	0.8543	0.8772	0.8686
ES-fixed*	T1	0.8223	0.9067	0.8930
ES-fixed*	T2	0.6770	0.8656	0.8377

Table 9: Post hoc sensitivity of pooled-IQM contrasts. Entries are nominal 95% bootstrap intervals. Anchor re-estimation recomputes the maximum within each draw; paired resampling keeps corresponding run seeds together across methods.

Comparison	Outcome	Re-estimated anchors	Paired seeds
OE minus PPO-DR	T2/T3 AUC	$[-0.6104, -0.5215]$	$[-0.6101, -0.5185]$
OE minus OE-fixed	T2/T3 AUC	$[-0.1800, -0.0678]$	$[-0.1810, -0.0692]$
PPO-DR minus PPO-fixed	T2/T3 AUC	$[-0.0006, 0.0196]$	$[0.0020, 0.0177]$
ES-DR minus ES-fixed	T2/T3 AUC	$[0.0204, 0.0385]$	$[0.0204, 0.0396]$
ES-DR minus ES-fixed	T1 zero-shot	$[0.0496, 0.0790]$	$[0.0523, 0.0764]$

Procedure	Suite	Zero-shot	Final	AUC
ES-fixed*	T3	0.8041	0.8530	0.8470

C Sensitivity analyses and population dynamics

Re-estimating the maximum anchor within each of 50,000 bootstrap draws and pairing corresponding run seeds leave the OE–PPO-DR and OE–OE-fixed intervals below zero and the ES-DR–ES-fixed intervals above zero (Table 9). For PPO-DR versus PPO-fixed, the paired IQM interval excludes zero, whereas the prespecified rank test on run means gives adjusted $p = 0.6232$. The paired comparison changes both the performance summary and the treatment of cross-method dependence.

Mean final adaptation counts per environment are approximately 20,000,010 for either OE variant, 20,017,188 for PPO-DR, 20,039,595 for PPO-fixed, 20,144,256 for ES-DR and 20,288,091 for ES-fixed. ES-fixed can overshoot its first two-million-step checkpoint by 49.5%; the largest checkpoint excess is 4.98% of the complete adaptation budget, and the largest final excess is 4.49%. Integrating linearly interpolated scores over actual step counts and truncating at 20 million steps gives an OE–PPO-DR difference of -0.5592 and an ES-DR–ES-fixed difference of 0.0289, with interval $[0.0206, 0.0388]$.

OE remains below PPO-DR when individual runs or environments are omitted and when raw competence replaces normalized scores. Figure 4 shows the performance distributions and pairwise run-mean comparisons.

We also compare raw competence with 90% of each run's mean T1 zero-shot score. OE and OE-fixed each reach that threshold in 43 of 200 T2/T3 run–environment pairs; PPO-DR, PPO-fixed and ES-DR each reach it in 40. A threshold set by T1 performance may be unattainable in a shifted environment.

Weight diversity is mean pairwise Euclidean distance among sixteen sampled neural parameter vectors; behavior diversity uses action distances on 256 fixed validation probes. Both measures are higher for OE than OE-fixed at the final checkpoint (Figure 5).

Three exploratory OE runs extend training to one billion steps (Figure 6). Their gains are -4.3657, 1.8893 and -0.9751, measured as the mean of the final two validation checkpoints minus

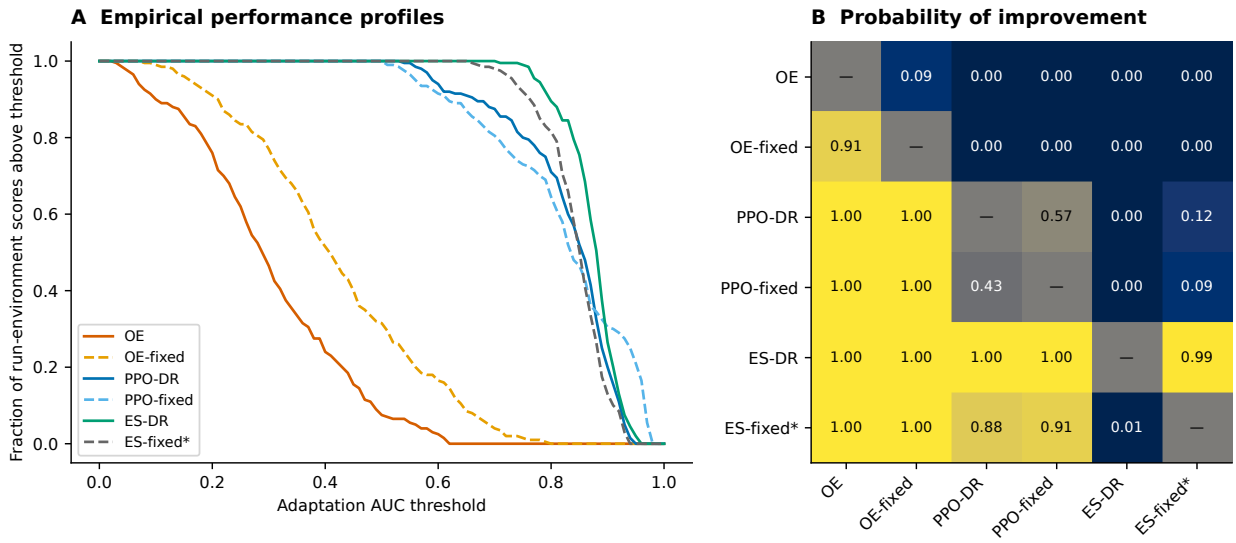


Figure 4: Descriptive performance summaries. Left: fraction above each AUC threshold among ten runs across twenty fixed environments. Right: fraction of the 100 cross-run comparisons in which the row method's mean exceeds the column method's mean, with ties counting one half.

the mean of the first two. The planned 4.3-point threshold was based on an unreliable estimate of variability.

D Supporting trait-only experiment

D.1 Design and analysis

The trait model combines inherited speed, size, sensing range and reach with fixed behavioral rules. The two phases contain 100 generations in environment A and 50 in B.

The groups are ET, the evolving population; F-food and F-pop, frozen copies of the food- and population-search winners; M-pop, mutating copies of the population-search winner; SV, a frozen initial population with standing variation; ET-freeze, the end-of-A evolved population with mutation disabled in B; ET-clone, copies of a randomly sampled end-of-A member with mutation disabled; and Oracle-B, the reference design selected directly in B. Oracle-B starts directly in B and is excluded from the confirmatory contrast family. Cloning changes the realized traits along with heterogeneity.

Each run seed defines its own searches and deployments, with 200 seeds per primary group and arm (2000–2199). Groups with the same seed share food, behavior and initial-position streams. The main B has 25 food items; three additional arms permit agent consumption in B, enlarge the B map to side 750, or reduce B's energy budget per generation to 25. We selected the larger-map and reduced-energy settings during pilot experiments before the corresponding final runs.

The generation rules derive from the MinuteLabs Evolution Simulator at commit `55af956` (Palfree, 2021). A has side length 500 and 50 uniformly placed food items. Main B has side length 300 and 25 items in five clusters with standard deviation 15; centers and food are redrawn each generation. A generation lasts at most 200 ticks, with an energy allowance of 500 per agent except in the reduced-energy B arm. Agents start at edge home positions, search using fixed

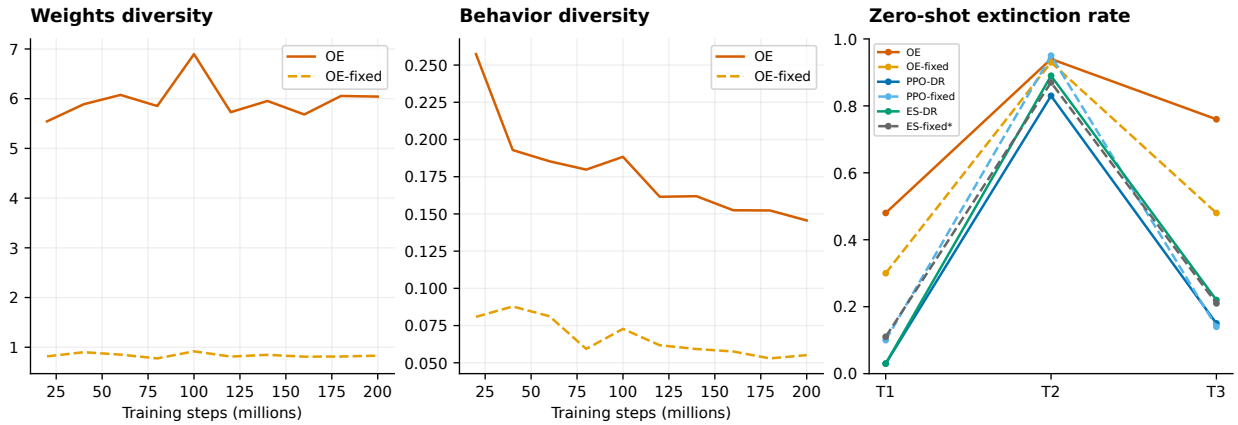


Figure 5: Population diagnostics and separate persistence. Weight and behavioral diversity summarize sixteen sampled agents per training population, averaged across ten runs. Right: extinction frequencies in the separate persistence assay without floor support.

steering rules and return home according to feeding and remaining energy. Exhaustion and, in the consumption arm, predation can kill agents; agents with no food at generation end also die. Survivors carry into the next generation, and each survivor that consumed at least two food units produces exactly one offspring. Food units count both food items and prey. Effective movement speed is $u = \text{speed} \times \text{size}/10$, and each active movement tick costs $10^{-4}(\text{size}^3 u^2 + \text{sense})$. Food supports survival and reproduction without replenishing this generation-level energy allowance. There is no externally supported population floor.

Initial populations have 50 agents. ET starts from speed/size/sense/reach values 10/10/20/1. Mutating offspring receive independent Gaussian trait perturbations with standard deviations 1/1/1/0.25; sensing range is clipped at zero and the other traits at the smallest positive normalized float64 value. ET and M-pop mutate in both phases. F-food, F-pop, SV and Oracle-B have mutation disabled throughout. ET-freeze and ET-clone are derived from ET after A and have mutation disabled in B. SV samples each trait independently and log-uniformly over the search ranges: speed and size 2.5–40, sense 5–80 and raw reach 0.25–4.

In the consumption arm, a predator can consume only an agent whose size is at most 80% of its own. Identical clones therefore cannot consume one another. Comparisons involving clones and heterogeneous populations also differ in access to this additional food source; differences among heterogeneous groups depend on their size distributions. Across all 50 B generations, mean prey counts per run are 150.8 for ET, 153.2 for M-pop, 18.4 for ET-freeze and 3.4 for SV, with zero for the four homogeneous frozen groups.

The A search begins with eight logarithmically spaced values for each trait. Merging designs with the same effective reach, $\max(\text{reach}, \text{size}/4)$, leaves 1,472 distinct designs. Stage one evaluates 50 clones of each design for 30 generations on shared streams. Stage two evaluates the top 32 designs per objective on eight fresh sets of streams. In stage two, population search scores mean population over generations 10–99; food search scores food per agent over generations 10–29, even when a candidate was simulated for longer. The comparison therefore changes both objective and scoring horizon. Search-selected mutating agents inherit effective reach.

P1 is population at the start of a generation, averaged over B generations 40–49 with zeros retained. P2 indicates that no agent would start B generation 50. Each arm has eight planned comparison pairs, except the consumption arm omits the population-search versus food-search pair

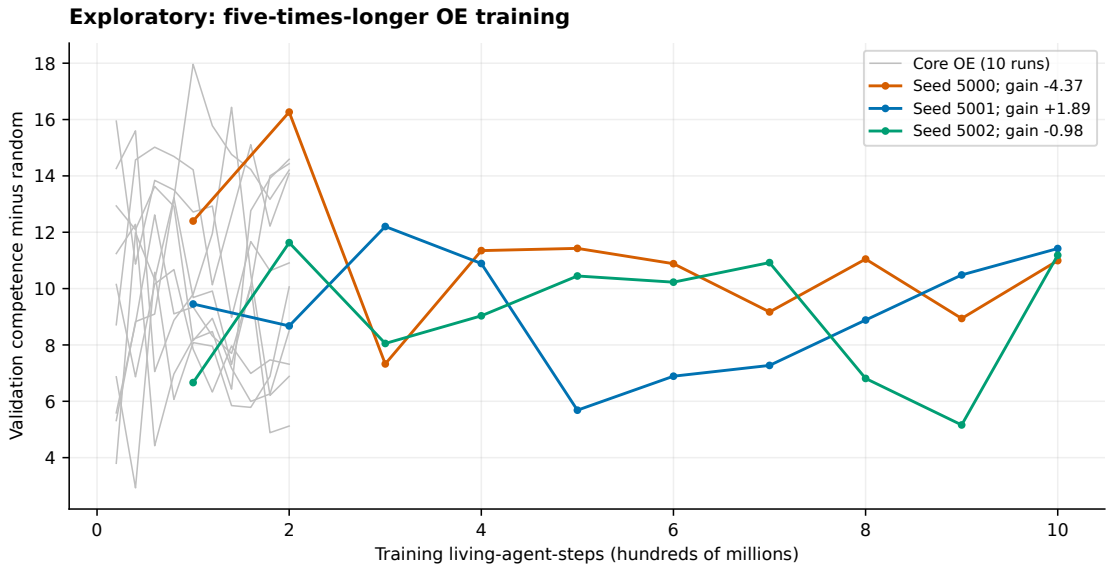


Figure 6: Exploratory longer training. Three fresh OE runs trained to one billion nominal steps, with the ten original 200-million-step runs in gray. Gain is the mean of the final two saved checkpoints minus the mean of the first two.

because those data repeat the main arm: 31 pairs in total. P1 uses a two-sided 10,000-permutation test of mean differences and a studentized bootstrap interval with 20,000 draws. P2 uses a two-sided Fisher test and a Newcombe interval for the difference in extinction proportions. Holm adjustment is within each arm's 16-test family, or 14 tests for the consumption arm. The tests treat groups as independent, although shared streams couple corresponding seeds.

The practical margins are 2.5 agents for P1 and 0.10 for P2. A "meaningfully different" label requires adjusted $p < 0.05$ and an absolute point estimate at least as large as the margin, even if the interval crosses it. Equivalence requires an interval at level $1 - 2(0.05)/k$, where k is the arm's family size, wholly inside the margins. Ceiling and floor frequencies show where boundary effects may mask population differences.

D.2 Results

In the main arm, F-pop reaches the population ceiling while ET is close to it. Their mean population difference is 0.6375 against a 2.5-agent margin, and neither group becomes extinct. F-pop also outperforms F-food (Figure 7). Tables 10 and 11 give the contrasts and group outcomes.

In the larger-map condition, F-pop exceeds ET by 5.74 agents (95% interval $[5.00, 6.66]$). Under reduced energy, ET-freeze has 3.93 fewer agents than ET (interval $[1.66, 6.12]$) and an extinction probability higher by 0.145 (interval $[0.047, 0.239]$).

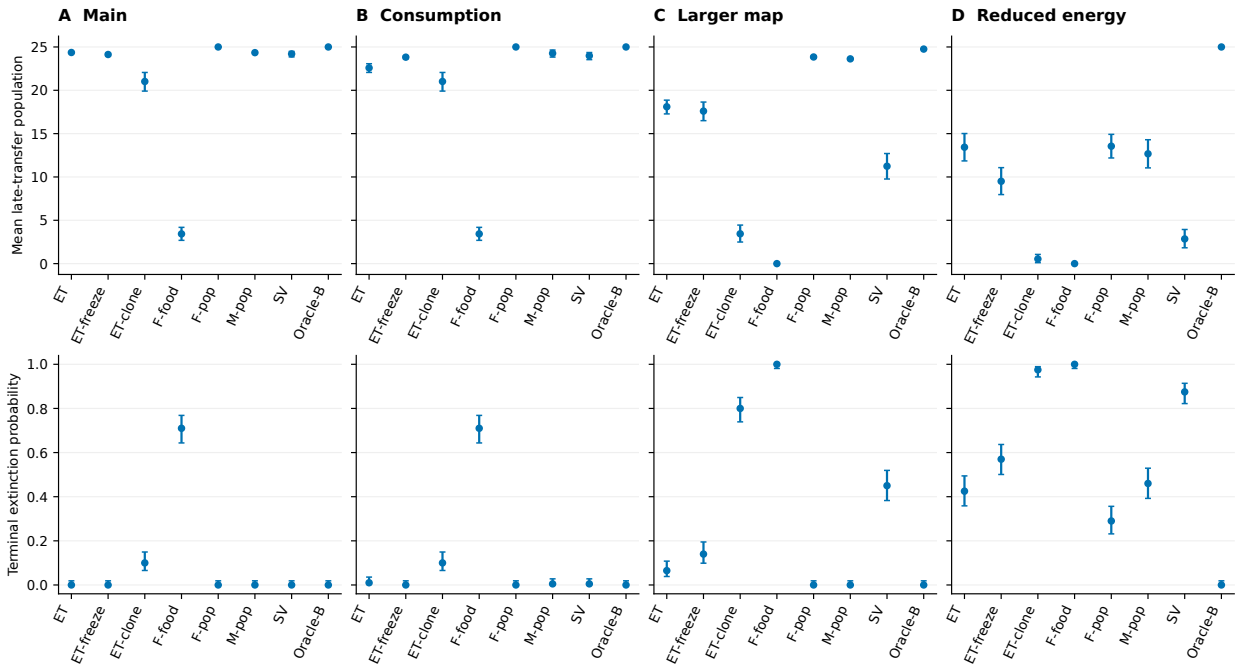


Figure 7: Supporting trait experiment. All eight primary groups in four arms, with 200 seeds per group. Upper panels: mean P1 and descriptive whole-seed percentile 95% intervals (20,000 draws). Lower panels: P2 proportions with Wilson 95% intervals. These differ from the studentized P1 and Newcombe P2 intervals for contrasts.

Table 10: Trait experiment comparisons. Differences are X minus Y, with pairs written X / Y. P1 is mean late-B population and P2 is terminal extinction probability. Entries are differences with 95% intervals. MD: significant with point estimate at or beyond the margin; EQ: equivalence under the specified interval rule; DU: detected difference with size unresolved; IN: inconclusive. Ceiling and floor diagnostics are in Table 11; full decision labels are in the accompanying CSV.

Arm	X / Y	P1 difference [CI]	P2 difference [CI]	Adjusted p (P1 / P2)	P1/P2
Main	F-food / ET	-20.93 $[-21.66, -20.11]$	$+0.71$ $[0.64, 0.77]$	0.0013 / 9.54×10^{-60}	MD/MD
Main	F-pop / ET	$+0.64$ $[0.59, 0.68]$	$+0.00$ $[-0.02, 0.02]$	0.0013 / 1	EQ/EQ
Main	M-pop / ET	-0.02 $[-0.08, 0.04]$	$+0.00$ $[-0.02, 0.02]$	1 / 1	EQ/EQ
Main	SV / ET	-0.16 $[-0.64, 0.10]$	$+0.00$ $[-0.02, 0.02]$	1 / 1	EQ/EQ
Main	ET-freeze / ET	-0.23 $[-0.41, -0.08]$	$+0.00$ $[-0.02, 0.02]$	0.04 / 1	EQ/EQ
Main	ET-clone / ET-freeze	-3.11 $[-4.36, -2.11]$	$+0.10$ $[0.06, 0.15]$	0.0013 / 1.62×10^{-5}	MD/MD
Main	F-pop / F-food	$+21.57$ $[20.76, 22.28]$	-0.71 $[-0.77, -0.64]$	0.0013 / 9.54×10^{-60}	MD/MD
Main	M-pop / F-pop	-0.66 $[-0.70, -0.62]$	$+0.00$ $[-0.02, 0.02]$	0.0013 / 1	EQ/EQ

Arm	X / Y	P1 difference [CI]	P2 difference [CI]	Adjusted p (P1 / P2)	P1/P2
Consumption	F-food / ET	-19.16 $[-20.02, -18.17]$	$+0.70$ $[0.63, 0.76]$	0.0012 / 5.18×10^{-56}	MD/MD
Consumption	F-pop / ET	$+2.41$ $[1.97, 3.02]$	-0.01 $[-0.04, 0.01]$	0.0012 / 1	DU/EQ
Consumption	M-pop / ET	$+1.68$ $[1.05, 2.34]$	-0.01 $[-0.03, 0.02]$	0.0012 / 1	DU/EQ
Consumption	SV / ET	$+1.40$ $[0.75, 2.05]$	-0.01 $[-0.03, 0.02]$	0.0012 / 1	EQ/EQ
Consumption	ET-freeze / ET	$+1.24$ $[0.76, 1.84]$	-0.01 $[-0.04, 0.01]$	0.0012 / 1	EQ/EQ
Consumption	ET-clone / ET-freeze	-2.80 $[-4.03, -1.81]$	$+0.10$ $[0.06, 0.15]$	0.0012 / 1.50×10^{-5}	MD/MD
Consumption	M-pop / F-pop	-0.73 $[-1.26, -0.38]$	$+0.01$ $[-0.01, 0.03]$	0.0018 / 1	EQ/EQ
Larger map	F-food / ET	-18.10 $[-18.83, -17.18]$	$+0.94$ $[0.89, 0.96]$	0.0012 / 5.98×10^{-98}	MD/MD
Larger map	F-pop / ET	$+5.74$ $[5.00, 6.66]$	-0.07 $[-0.11, -0.03]$	0.0012 / 0.0012	MD/DU
Larger map	M-pop / ET	$+5.53$ $[4.80, 6.43]$	-0.07 $[-0.11, -0.03]$	0.0012 / 0.0012	MD/DU
Larger map	SV / ET	-6.86 $[-8.56, -5.13]$	$+0.39$ $[0.31, 0.46]$	0.0012 / 1.48×10^{-18}	MD/MD
Larger map	ET-freeze / ET	-0.50 $[-1.87, 0.81]$	$+0.07$ $[0.02, 0.14]$	0.9333 / 0.07	EQ/IN
Larger map	ET-clone / ET-freeze	-14.16 $[-15.54, -12.58]$	$+0.66$ $[0.58, 0.72]$	0.0012 / 1.44×10^{-41}	MD/MD
Larger map	F-pop / F-food	$+23.84$ $[23.70, 23.97]$	-1.00 $[-1.00, -0.97]$	0.0012 / 3.11×10^{-118}	MD/MD
Larger map	M-pop / F-pop	-0.22 $[-0.39, -0.04]$	$+0.00$ $[-0.02, 0.02]$	0.07 / 1	EQ/EQ
Reduced energy	F-food / ET	-13.43 $[-15.02, -11.81]$	$+0.57$ $[0.50, 0.64]$	0.0012 / 3.99×10^{-44}	MD/MD
Reduced energy	F-pop / ET	$+0.12$ $[-1.98, 2.27]$	-0.14 $[-0.23, -0.04]$	1 / 0.0329	IN/MD
Reduced energy	M-pop / ET	-0.76 $[-3.04, 1.49]$	$+0.04$ $[-0.06, 0.13]$	1 / 1	IN/IN
Reduced energy	SV / ET	-10.58 $[-12.43, -8.63]$	$+0.45$ $[0.36, 0.53]$	0.0012 / 1.01×10^{-20}	MD/MD
Reduced energy	ET-freeze / ET	-3.93 $[-6.12, -1.66]$	$+0.14$ $[0.05, 0.24]$	0.0032 / 0.0303	MD/MD
Reduced energy	ET-clone / ET-freeze	-8.97 $[-10.63, -7.33]$	$+0.41$ $[0.33, 0.48]$	0.0012 / 1.39×10^{-23}	MD/MD
Reduced energy	F-pop / F-food	$+13.55$ $[12.15, 14.93]$	-0.71 $[-0.77, -0.64]$	0.0012 / 9.54×10^{-60}	MD/MD
Reduced energy	M-pop / F-pop	-0.88 $[-3.02, 1.24]$	$+0.17$ $[0.08, 0.26]$	1 / 0.0044	IN/MD

Table 11: Trait-study group outcomes and boundary diagnostics. All groups have 200 seeds. High/low are fractions with P1 at least 22.5 / at most 2.5. In the consumption arm, 25 is not a hard population ceiling. A-dead identifies losses preceding the environmental shift.

Arm	Group	Mean P1	Extinct	A-dead	High	Low
Main	F-food	3.43	142/200	62	0.00	0.69
Main	ET	24.36	0/200	0	1.00	0.00
Main	F-pop	25.00	0/200	0	1.00	0.00
Main	M-pop	24.34	0/200	0	1.00	0.00
Main	SV	24.20	0/200	0	0.92	0.00
Main	ET-freeze	24.13	0/200	0	0.90	0.00
Main	ET-clone	21.02	20/200	0	0.73	0.10
Main	Oracle-B	25.00	0/200	0	1.00	0.00
Consumption	F-food	3.43	142/200	62	0.00	0.69
Consumption	ET	22.59	2/200	0	0.61	0.01
Consumption	F-pop	25.00	0/200	0	1.00	0.00
Consumption	M-pop	24.27	1/200	0	0.85	0.01
Consumption	SV	23.99	1/200	0	0.90	0.01
Consumption	ET-freeze	23.82	0/200	0	0.86	0.00
Consumption	ET-clone	21.02	20/200	0	0.73	0.10
Consumption	Oracle-B	25.00	0/200	0	1.00	0.00
Larger map	F-food	0.00	200/200	62	0.00	1.00
Larger map	ET	18.10	13/200	0	0.10	0.07
Larger map	F-pop	23.84	0/200	0	0.93	0.00
Larger map	M-pop	23.63	0/200	0	0.94	0.00
Larger map	SV	11.24	90/200	0	0.24	0.45
Larger map	ET-freeze	17.60	28/200	0	0.33	0.14
Larger map	ET-clone	3.44	160/200	0	0.04	0.79
Larger map	Oracle-B	24.76	0/200	0	1.00	0.00
Reduced energy	F-food	0.00	200/200	62	0.00	1.00
Reduced energy	ET	13.43	85/200	0	0.50	0.42
Reduced energy	F-pop	13.55	58/200	0	0.34	0.28
Reduced energy	M-pop	12.67	92/200	0	0.54	0.46
Reduced energy	SV	2.85	175/200	0	0.09	0.88
Reduced energy	ET-freeze	9.50	114/200	0	0.28	0.56
Reduced energy	ET-clone	0.54	195/200	0	0.01	0.97
Reduced energy	Oracle-B	25.00	0/200	0	1.00	0.00

F-food loses 62 of 200 main-arm runs during A; these runs contribute zeros to B outcomes. Restricting the comparison to populations that reach B would select survivors. The no-move control continues ET's end-of-A population for fifty further generations in A. Mean late population is 48.0675 with mutation, 47.9895 after freezing, and 40.9730 after cloning. The freeze-minus-mutation difference is -0.0780, interval $[-0.2720, 0.1140]$; the clone-minus-freeze difference is -7.0165, interval $[-9.1271, -5.0470]$. Cloning therefore reduces population even without an environmental shift.

Searching directly in B improves population score by less than 2.5 agents over searching in A in 200 of 200 main-arm runs, 200 of 200 consumption-arm runs, 199 of 200 larger-map runs and 67 of

200 reduced-energy runs. The standing-variation group retains only one design at transfer in 153 of 200 runs, limiting its use as a diverse-population control.

In an exploratory 100-seed subset, starting ET directly in B instead of transferring from A changes P1 by -0.22 in the main arm, -17.62 in the larger-map arm and -12.85 in the reduced-energy arm.

E Protocols and reproducibility

The protocols were written before the corresponding runs but were not publicly registered. A three-seed pilot informed the choice of ten final runs per configuration.

The neural experiments use code revision `16e8657` for the five initial configurations and `06ff7b2` for ES-fixed. The trait experiment uses `a1e1cdc` for simulation and `561bcd4` for confirmatory analysis.

The initial neural runs used NVIDIA RTX 4090 hardware, JAX 0.11.2 and Python 3.13.18. Statistical reanalysis used Python 3.12.14 on CPU.

The released data contain sixty final neural evaluations, 19,800 adaptation checkpoint observations, trait trajectories, final policies, training snapshots and tuning results. Policies from the initial tuning runs were not retained. `REPRODUCE.md` gives the commands for the analyses, tables and figures.